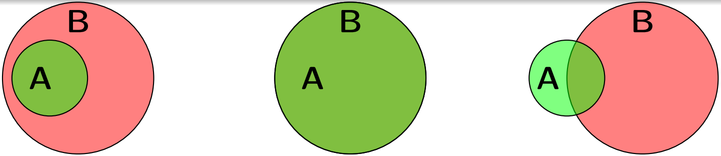
Section 7.2: Venn Diagrams

Finite Mathematics MTH 132 Spring 2014

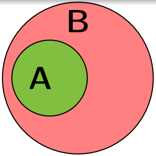
Dr. James Ashe

Johnson C. Smith University

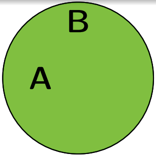
Venn diagrams

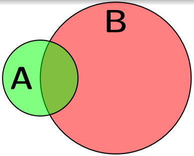


Venn Diagrams



$A \subseteq B$ and $A \subset B$





Venn Diagrams

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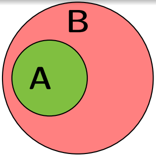

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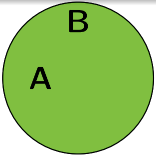

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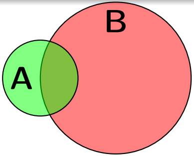

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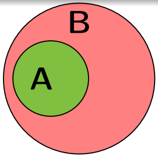

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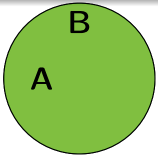

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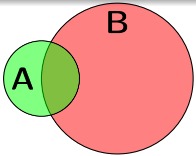
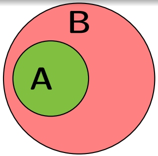


$A \not\subset B$

Venn Diagrams



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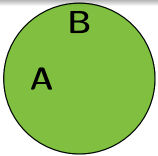

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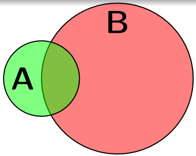


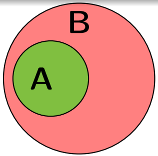
$A \not\subseteq B$

Venn Diagrams

Problem 7.2.1. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$$A \_\_\_ E$$

Venn diagrams



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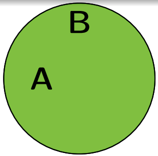

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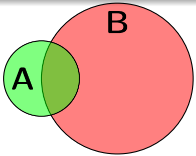

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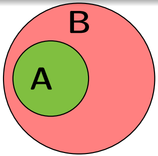
Venn Diagrams

Problem 7.2.2. Let $A = \{2,4,6,10\}$, $B = \{2,4,8,10\}$, $C = \{4,8,12\}$, $D = \{2,10\}$, $E = \{6\}$, and $U = \{2,4,6,8,10,12,14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$$\emptyset\_\_\_A$$

Venn diagrams



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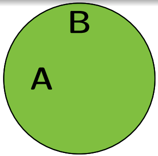

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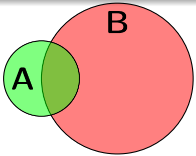

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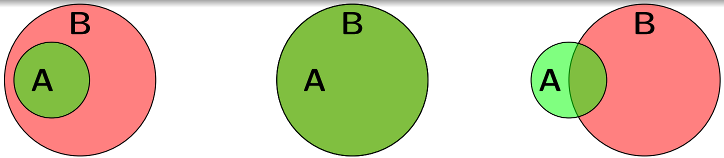
Venn Diagrams

Problem 7.2.3. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$$A \_\_\_ U$$

Venn diagrams



$A \subseteq B$ and $A \subset B$ $\quad$ $A \subseteq B$ $\quad$ $A \not\subseteq B$

Venn Diagrams

Problem 7.2.4. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$$A\underline{\quad}U$$

Set Operations

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Union

The *union* of A and B is the set of all elements in A or B

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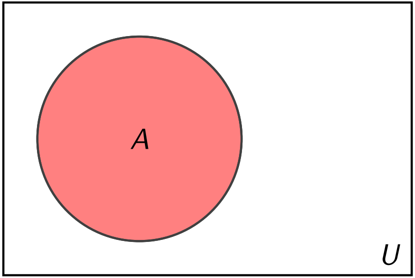
- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Compliment

The *compliment* of A is the set of everything in U but not in A.

- $A' = \{x \mid x \notin A \text{ and } x \in U\}$



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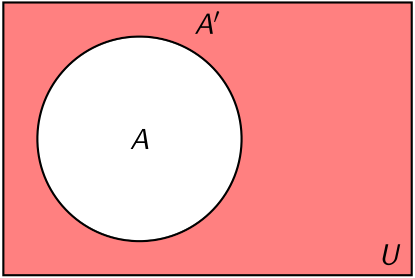

Set Operation

Problem 7.2.5. Let $U = \{1,2,3,4,5,6,7,8,9\}$, $X = \{2,4,6,8\}$, $Y = \{2,3,4,5,6\}$, and $Z = \{1,2,3,8,9\}$. List the members of each set, using set braces.

$X \cup Y$

Problem 7.2.6. Let $U = \{1,2,3,4,5,6,7,8,9\}$, $X = \{2,4,6,8\}$, $Y = \{2,3,4,5,6\}$, and $Z = \{1,2,3,8,9\}$. List the members of each set, using set braces.

$X \cap Y$

Set Operation

Problem 7.2.7. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $X = \{2, 4, 6, 8\}$, $Y = \{2, 3, 4, 5, 6\}$, and $Z = \{1, 2, 3, 8, 9\}$. List the members of each set, using set braces.

X'

Problem 7.2.8. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $X = \{2, 4, 6, 8\}$, $Y = \{2, 3, 4, 5, 6\}$, and $Z = \{1, 2, 3, 8, 9\}$. List the members of each set, using set braces.

$X' \cap Y'$

Set Operation

Problem 7.2.9. Let $U = \{1,2,3,4,5,6,7,8,9\}$, $X = \{2,4,6,8\}$, $Y = \{2,3,4,5,6\}$, and $Z = \{1,2,3,8,9\}$. List the members of each set, using set braces.

$X' \cap (Y' \cup Z)$

Problem 7.2.10. Let $U = \{1,2,3,4,5,6,7,8,9\}$, $X = \{2,4,6,8\}$, $Y = \{2,3,4,5,6\}$, and $Z = \{1,2,3,8,9\}$. List the members of each set, using set braces.

$(X \cap Y) \cup (X \cap Z)$

Problem 7.2.11. Let $U = \{1,2,3,4,5,6,7,8,9\}$, $X = \{2,4,6,8\}$, $Y = \{2,3,4,5,6\}$, and $Z = \{1,2,3,8,9\}$. List the members of each set, using set braces.

$(X \cap Y) \cap \{1,7\}$

- How many do not have a DVD player?

Example 7.2.1.

A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

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Venn Diagram Applications

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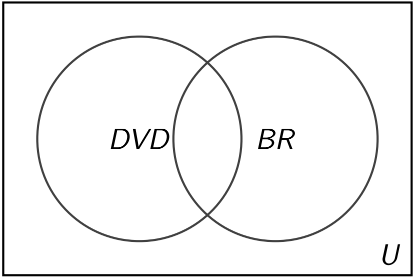
A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- How many do not have a DVD player?
- How many have neither a DVD or Blue Ray player?



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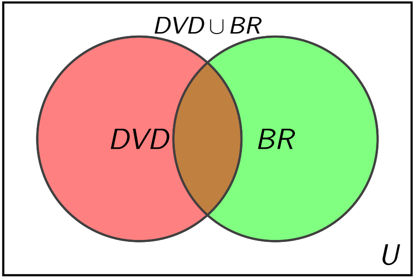

Union Rule

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

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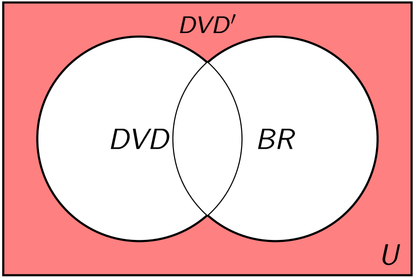

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Venn Diagram Applications

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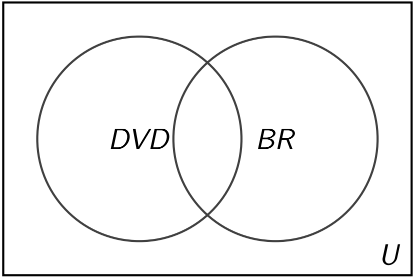
- How many have neither a DVD or Blue Ray player?
- How many have a Blue Ray but not a DVD player?



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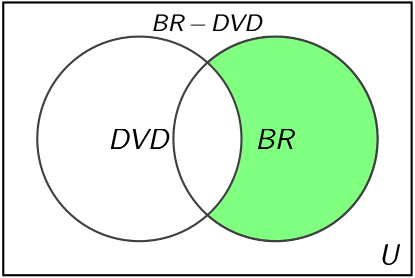

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a. How many students are not math majors?

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a. How many students are not math majors?
b. How many students are both math majors and female?

Venn Diagram Applications

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Venn Diagram Applications

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Venn Diagram Applications

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Venn Diagram Applications

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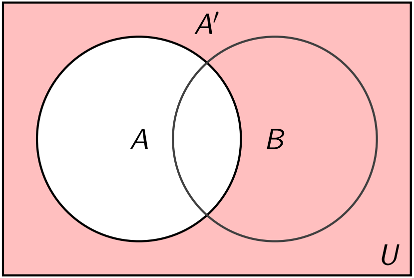

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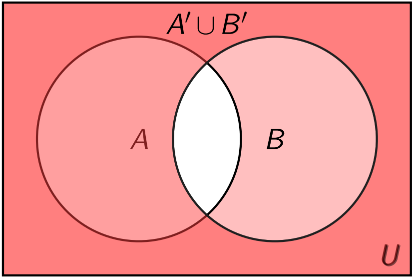

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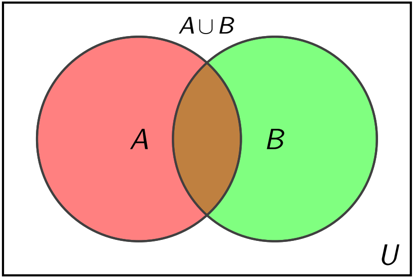

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